

NuSTAR Calibration Coffee Hour

MLI Updates

Hannah Earnshaw, 2025/03/27

What is MLI?

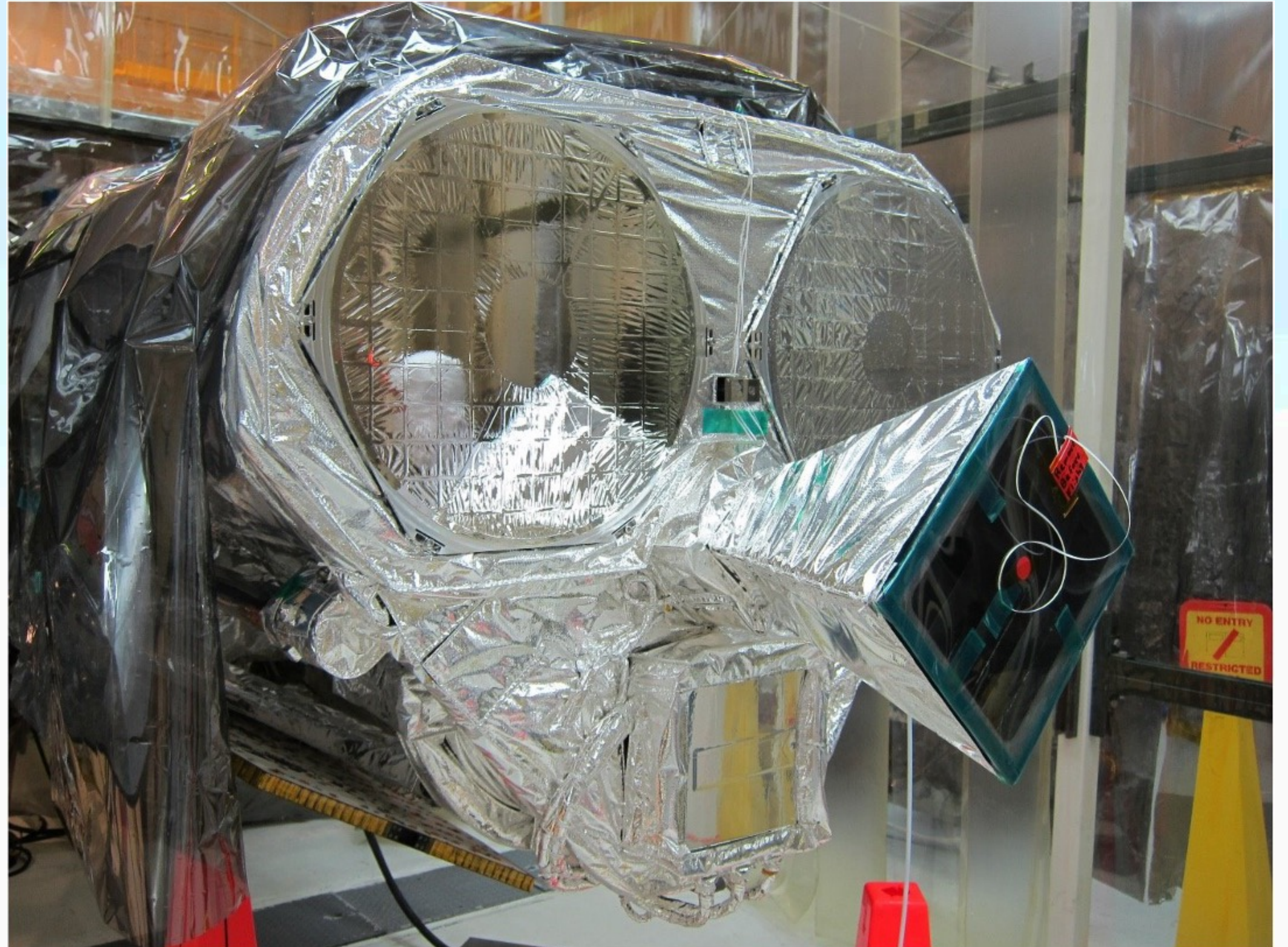
MLI = Multi-layer Insulation

Helps keep the NuSTAR optics thermally stable

Photo is NuSTAR pre-launch, with OPT0 closest to us and OPT1 to the right

OPT0 serves focal plane module FPMA, and OPT1 serves FPMB

Mostly transparent to X-rays, but there is some energy-dependent attenuation at low energies



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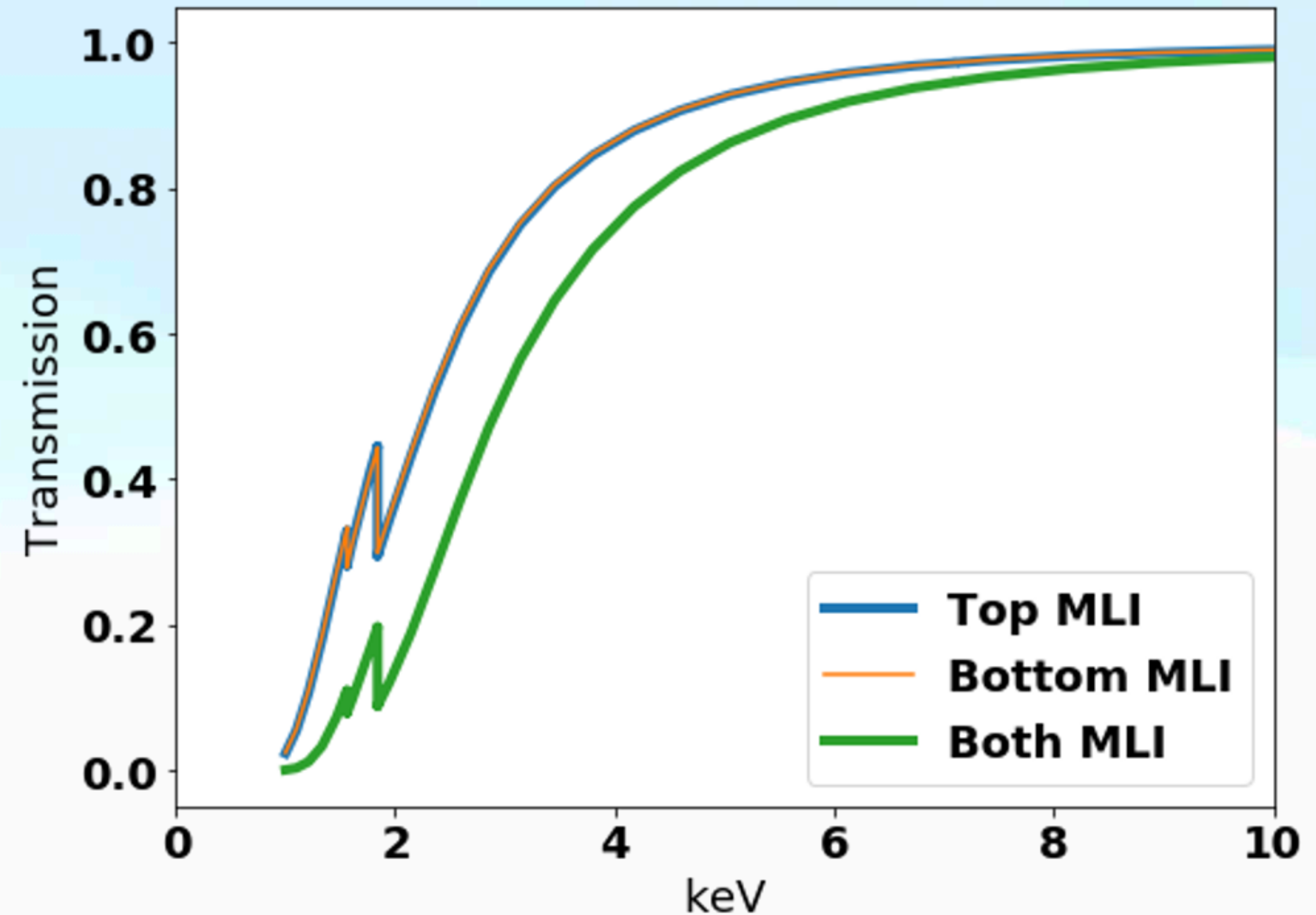
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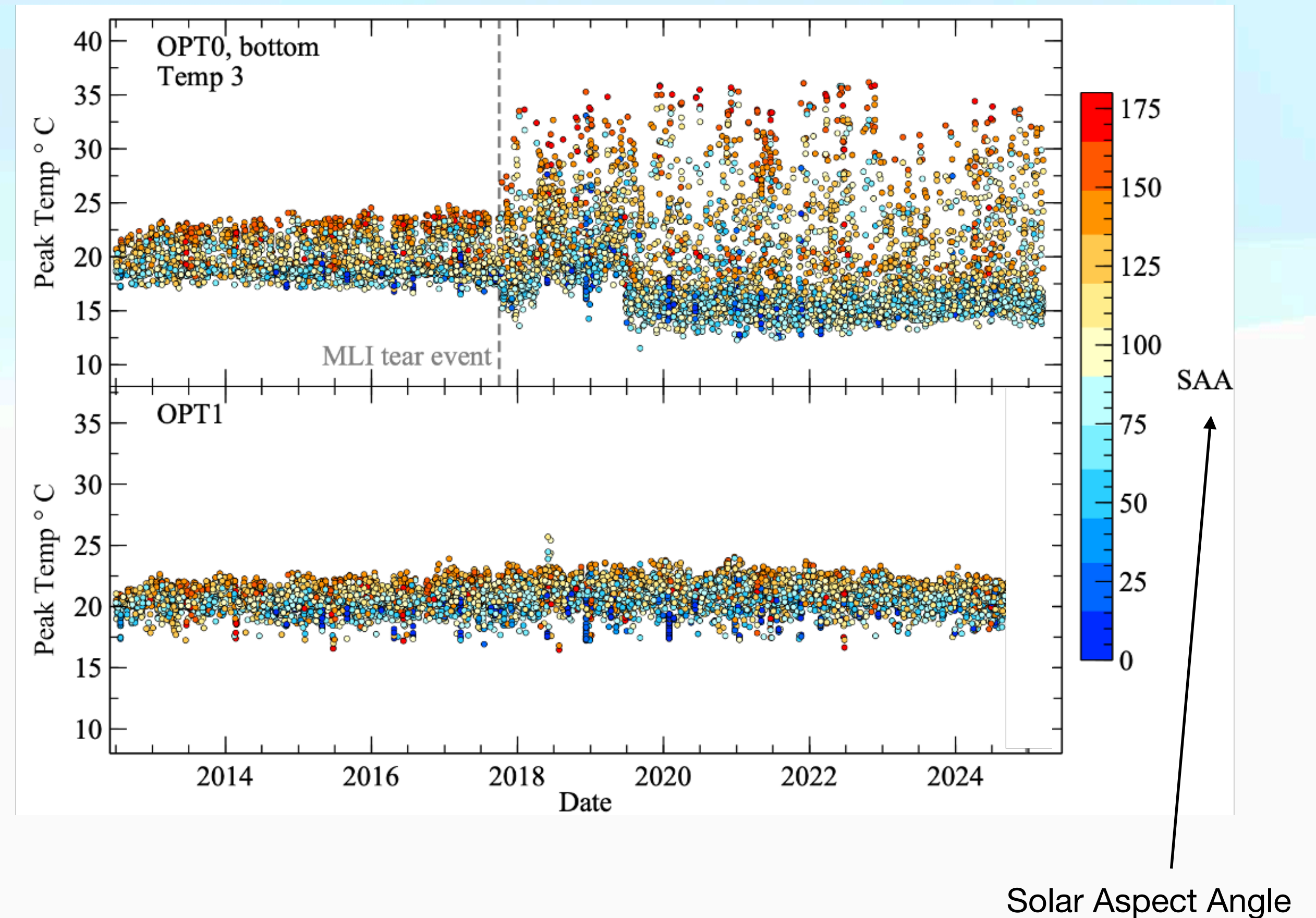
Finding a tear using temperature monitoring

Optics have six thermistors on each module, three on top and three on the bottom

Optics normally held at steady temperature of ~20C

A sudden change in the temperature indicates that something has happened! Some parts are getting directly illuminated by the Sun, others are losing heat into space.

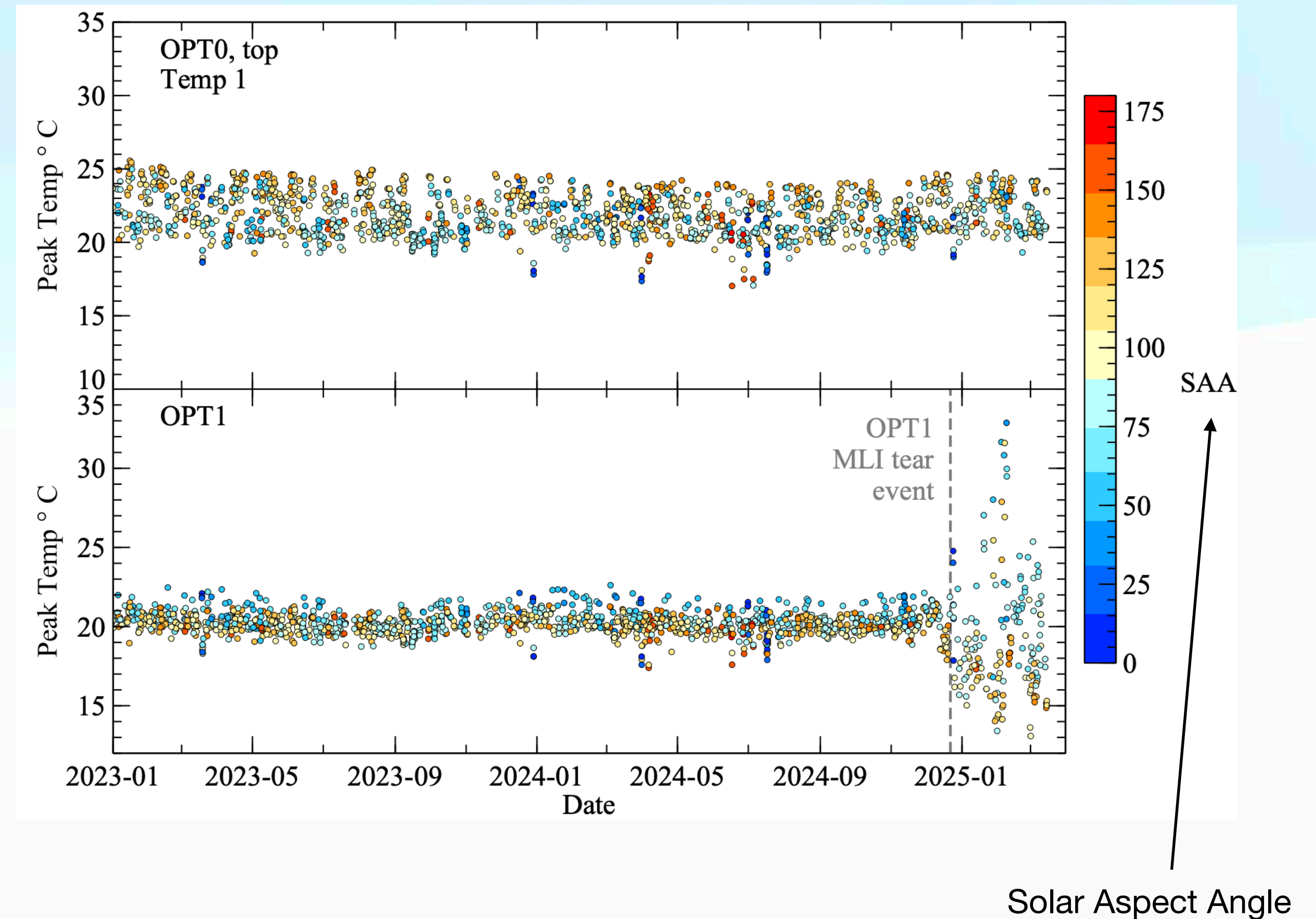
First tear event on OPT0 in late 2017 led to high temperatures measured by Temp 3, where optic was illuminated at high angles from the Sun



Finding a tear using temperature monitoring

In late Dec 2024, we found a new tear through noticing a change in OPT1 temperatures

Since this is the top of the optic, it receives more illumination (-> higher temperatures) when the spacecraft is pointing near the Sun

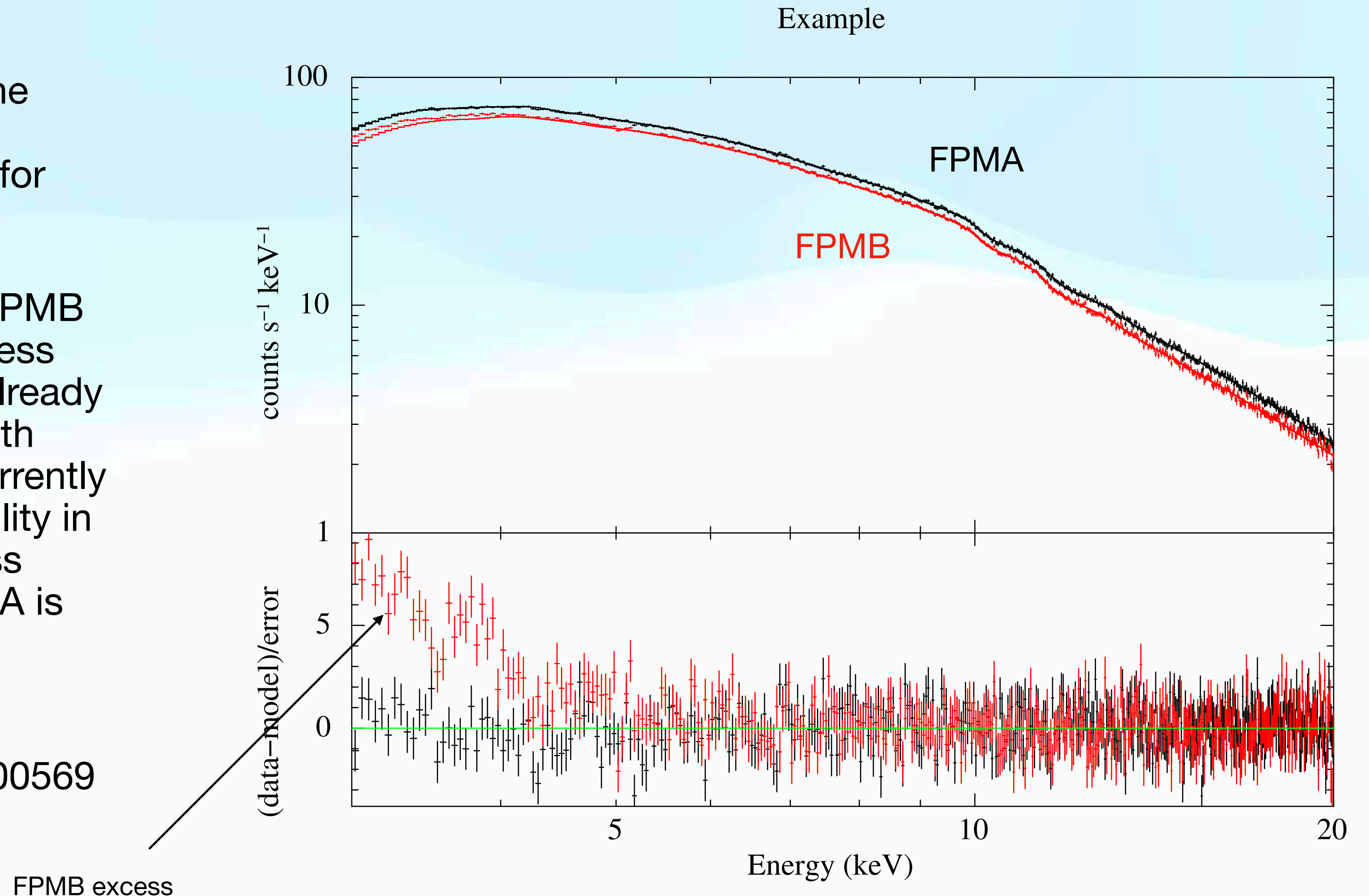


Effect on science

Scheduling impacts: for the moment, we have paused observations with $SAA < 42$ for safety purposes

Data impacts: increased FPMB throughput leads to an excess over FPMA (FPMA tear is already accounted for in CALDB with ARF file updates). There currently appears to be some instability in the amount of FPMB excess throughput (thankfully FPMA is stable!)

More info: Madsen (2020)
<https://arxiv.org/abs/2005.00569>



What to do if your data is affected

CAVEAT

The following **only** applies to...

- **Bright** sources ($>\sim 100$ c/s) that were...
- Observed **after** 2024 which show...
- A clear discrepancy between FPMA and FPMB (increased FPMB) at energies < 7 keV

If you see this, let us know! We can help to confirm that MLI is the source of the discrepancy and advise on analysis

Obtain the MLI mtable model from the NuSTAR Operations Site here: https://nustarsoc.caltech.edu/NuSTAR_Public/NuSTAROperationSite/mli.php

Available MLI models:

nuMLIv1.mod

Generated May 01, 2020

What to do if your data is affected

How to use the MLI mtable:

1. First, ignore FPMB between 3 and 7 keV when doing your original fits. Make sure the model includes a constant offset between FPMA and FPMB. Find the value of the constant and freeze it. Possibly also freeze NH, since it can be degenerate with MLI fraction.
2. Notice FPMB between 3 and 7 keV and add the mtable to the model. Set the value of the MLI fraction to 1.0 for FPMA and freeze it. Allow only the MLI fraction of FPMB to fit.
3. Please note: The range of MLI fraction may be between 0.6 and 1.0. If you have any concerns about a particular observation, please contact the NuSTAR Helpdesk at the HEASARC.

```
XSPEC12>data 1:1 nuFPMAspec_grp.pha 2:2 nuFPMBSpec_grp.pha
XSPEC12>ignore bad
XSPEC12>ignore **0.-3. 79.-**
XSPEC12>ignore 2:3.-7.
```

```
XSPEC12>model const*tbabs*powerlaw
XSPEC12>fit
XSPEC12>freeze 2,5
```

=====						
Model		constant<1>*TBabs<2>*powerlaw<3>	Source No.: 1	Active/On		
Model	Model	Component	Parameter	Unit	Value	
par	comp					
Data group: 1						
1	1	constant	factor		1.00000	frozen
2	2	TBabs	nH	10^22	0.735897	frozen
3	3	powerlaw	PhoIndex		2.10713	+/- 9.38088E-04
4	3	powerlaw	norm		9.55044	+/- 1.92313E-02
Data group: 2						
5	1	constant	factor		0.978625	frozen
6	2	TBabs	nH	10^22	0.735897	= p2
7	3	powerlaw	PhoIndex		2.10713	= p3
8	3	powerlaw	norm		9.55044	= p4

```
XSPEC12>notice 2:3.-7.
XSPEC12>editmod const*tbabs*powerlaw*mtable{nuMLIv1.mod}
```

Input parameter value, delta, min, bot, top, and max values for ...						
0.8	0.005(	0.008)	0	0	1	1
5:data group 1::fractionalMLI:MLI2_frac>1 -1						
Input parameter value, delta, min, bot, top, and max values for ...						
0.8	0.005(	0.008)	0	0	1	1
10:data group 2::fractionalMLI:MLI2_frac>1						

```
XSPEC12>fit
```

=====						
Model		constant<1>*TBabs<2>*powerlaw<3>*mtable{nuMLIv1.mod}<4>	Source No.: 1	Active/On		
Model	Model	Component	Parameter	Unit	Value	
par	comp					
Data group: 1						
1	1	constant	factor		1.00000	frozen
2	2	TBabs	nH	10^22	0.735897	frozen
3	3	powerlaw	PhoIndex		2.10676	+/- 9.22952E-04
4	3	powerlaw	norm		9.53066	+/- 1.87343E-02
5	4	fractionalMLI	MLI2_frac		1.00000	frozen
Data group: 2						
6	1	constant	factor		0.978625	frozen
7	2	TBabs	nH	10^22	0.735897	= p2
8	3	powerlaw	PhoIndex		2.10676	= p3
9	3	powerlaw	norm		9.53066	= p4
10	4	fractionalMLI	MLI2_frac		0.813340	+/- 7.34013E-03

Going forwards

We are continuing to investigate thermal behavior as telescope is pointed towards the Sun – we hope to be able to resume near-Sun observations in the future

Without OPT1 as a stable baseline, modeling the evolution of the MLI in both optics at once is a little more difficult (but not impossible!) We will be monitoring the Crab and other bright sources more regularly

We plan to release a CALDB update for the FPMB ARF later this year. As the tear is currently unstable, MLI fitting may still be required in a small number of cases

Calibration Coffee slides available at: https://github.com/NuSTAR/calibration_coffee